Angular Momentum Representation For Free particle

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Abstract

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Keywords: a

1 Problem Setting

We seek a complete formulation of the angular-momentum representation for a non-relativistic free particle. The central objective is the diagonalization of the Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} \tag{1}$$

within the basis defined by the Complete Set of Commuting Observables (CSCO): $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$. This corresponds to solving the time-dependent Schrödinger equation (TDSE) by exploiting the isotropy of free space.

The formulation is motivated by three theoretical necessities:

1. **Symmetry Realization:** To represent the rotation group SO(3) via its generators acting on the Hilbert space of the free particle.
2. **Conservation Laws:** To explicitly link the invariance of $\hat{H}$ under spatial rotations to the conservation of angular momentum.
3. **Separation of Variables:** To derive the natural basis for spherical geometries, generalizing to central potential problems.

1.1 Physical Scenario

Consider a spinless particle of mass m in Euclidean space $\mathbb{R}^3$. The absence of external potentials implies the Hamiltonian is purely kinetic. The isotropy of the system ensures that the generator of rotations, the angular momentum vector operator $\hat{\mathbf{L}}$, commutes with the Hamiltonian:

$$[\hat{H}, \hat{\mathbf{L}}] = 0. \tag{2}$$

Consequently, stationary states may be labeled simultaneously by energy (E_k), total angular momentum (ℓ), and its projection (m), i.e., $\hat{H}$, $\hat{L}^2$ and $\hat{L}_z$ admit simultaneous eigenstates.

1.2 Construction Objectives

The project proceeds through the following formal steps:

- **Algebraic Definition:** Derivation of the operator $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ utilizing fundamental canonical commutation relations (CCR).
- **Lie Algebra $\mathfrak{so}(3)$:** Verification of the closure relation $[\hat{L}_i, \hat{L}_j] = i\hbar\,\epsilon_{ijk}\hat{L}_k$, establishing the connection to the structure constants of the rotation group.
- **Spectral Analysis:** Determination of the spectrum for $\hat{L}^2$ and $\hat{L}_z$, yielding the spherical harmonics $Y_{\ell m}(\theta, \phi)$ as coordinate representations of the angular eigenbasis.
- **Basis Transformation:** Construction of the unitary transformation connecting the linear momentum basis $|\mathbf{k}\rangle$ to the spherical basis $|k, \ell, m\rangle$.

1.3 Mathematical Prerequisites

The derivation assumes fluency in:

- **Abstract Formalism:** Hilbert space $\mathcal{H}$, Dirac notation, and spectral theory of Hermitian operators.
- **Symplectic Structure:** Canonical commutators $[\hat{x}_i, \hat{p}_j] = i\hbar\,\delta_{ij}$.
- **Integral Transforms:** Fourier analysis relating position ($\mathbf{r}$) and momentum ($\mathbf{p}$) representations.
- **Special Functions:** Properties of Spherical Harmonics and Spherical Bessel functions.

1.4 Formal Outcome

The construction culminates in the general solution to the TDSE, expressed as a superposition of spherical waves:

$$|\psi(t)\rangle = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} \mathrm{d}k\, k^2\, \psi_{k\ell m}(0)\, e^{-i\frac{\hbar k^2}{2m}t}\, |k, \ell, m\rangle\,. \tag{3}$$

This expansion connects the abstract eigenkets to the coordinate representation via the spherical Bessel transform.

2 Starting Point: Fundamental Formalism

We formally define the Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3)$ and the dynamical structure of the free particle. The following definitions establish the algebraic and spectral foundations required for the angular-momentum decomposition.

2.1 Dynamics and Unitary Evolution

The dynamics are governed by the self-adjoint Hamiltonian operator $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m}$. The state vector $|\psi(t)\rangle$ evolves deterministically from an initial state $|\psi_0\rangle$ via the strongly continuous unitary group $\hat{U}(t)$:

$$|\psi(t)\rangle = \hat{U}(t)\,|\psi_0\rangle = \exp\left(-\frac{i}{\hbar}\hat{H}t\right)|\psi_0\rangle\,. \tag{4}$$

This abstract evolution implies the Schrödinger equation $i\hbar\partial_t\,|\psi\rangle = \hat{H}\,|\psi\rangle$ via Stone's Theorem.

2.2 Continuous Representations

We utilize two continuous bases, position $\{|\mathbf{r}\rangle\}$ and linear momentum $\{|\mathbf{p}\rangle\}$, normalized to the Dirac distribution.

2.2.1 Basis Definitions and Orthogonality

The basis vectors satisfy the eigenvalue equations and completeness relations (Resolution of Identity $\hat{\mathbb{I}}$):

$$\hat{\mathbf{r}}\,|\mathbf{r}\rangle = \mathbf{r}\,|\mathbf{r}\rangle\,, \qquad \int_{\mathbb{R}^3} \mathrm{d}^3r\;|\mathbf{r}\rangle\langle\mathbf{r}| = \hat{\mathbb{I}}, \tag{5}$$

$$\hat{\mathbf{p}}\,|\mathbf{p}\rangle = \mathbf{p}\,|\mathbf{p}\rangle\,, \qquad \int_{\mathbb{R}^3} \mathrm{d}^3p\;|\mathbf{p}\rangle\langle\mathbf{p}| = \hat{\mathbb{I}}. \tag{6}$$

The inner product defines the transformation kernel (Fourier kernel) between the canonically conjugate variables:

$$\langle\mathbf{r}\rangle\,\mathbf{p} = \frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}\right). \tag{7}$$

2.2.2 Wavefunction Representations

The projections of the state vector yield the coordinate and momentum space wavefunctions:

- **Coordinate Space:** $\psi(\mathbf{r},t) \equiv \langle\mathbf{r}\rangle\,\psi(t)$. The momentum operator acts as the generator of spatial translations:

$$\langle\mathbf{r}|\,\hat{\mathbf{p}}\,|\psi\rangle = -i\hbar\nabla\psi(\mathbf{r}). \tag{8}$$

- **Momentum Space:** $\tilde{\psi}(\mathbf{p},t) \equiv \langle \mathbf{p} \rangle \, \psi(t)$. Since $\hat{H}$ is diagonal in this basis, the time evolution is strictly a phase modulation:

$$\tilde{\psi}(\mathbf{p},t) = \exp\left(-\frac{i}{\hbar}\frac{p^2}{2m}t\right)\tilde{\psi}(\mathbf{p},0). \tag{9}$$

2.3 Spectral Decomposition and Density of States

For the free particle, the energy E is degenerate. The transition from momentum integration to energy integration requires the spectral density function. Using the dispersion relation $E_p = p^2/2m$:

$$\int \mathrm{d}^3 p = \int_0^\infty p^2 \mathrm{d}p \int_{S^2} \mathrm{d}\Omega. \tag{10}$$

Changing variables from p to E introduces the density of states factor via the identity:

$$\mathrm{d}p = \frac{\mathrm{d}p}{\mathrm{d}E}\mathrm{d}E = \frac{m}{p}\mathrm{d}E = \sqrt{\frac{m}{2E}}\mathrm{d}E. \tag{11}$$

This transformation is critical for isolating the radial (energy) dependence from the angular dependence in the final construction.

2.4 Symmetry Generators

The continuous symmetries of the system are generated by Hermitian operators, implying specific conservation laws via Ehrenfest's Theorem, $\frac{\mathrm{d}}{\mathrm{d}t}\langle \hat{A} \rangle = \frac{1}{i\hbar}\langle [\hat{A},\hat{H}] \rangle$.

1. **Spatial Translations:** Generated by $\hat{\mathbf{p}}$. Since $[\hat{\mathbf{p}},\hat{H}] = 0$, linear momentum is conserved.

$$\hat{T}(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right). \tag{12}$$

2. **Galilean Boosts:** Generated by $\hat{\mathbf{r}}$ (up to mass scaling). This shifts the momentum spectrum:

$$\hat{T}_{\mathbf{p}}(\mathbf{q}) = \exp\left(\frac{i}{\hbar}\mathbf{q}\cdot\hat{\mathbf{r}}\right). \tag{13}$$

3 The Angular Momentum Operator

In classical mechanics, the angular momentum of a particle with position vector $\mathbf{r}$ and linear momentum $\mathbf{p}$ is defined as the vector product $\mathbf{L}_{cl} = \mathbf{r} \times \mathbf{p}$. To transition to quantum mechanics, we apply the correspondence principle, promoting the dynamical variables to linear operators acting on the Hilbert space: $\mathbf{r} \to \hat{\mathbf{r}}$ and $\mathbf{p} \to \hat{\mathbf{p}}$.

However, since $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are non-commuting operators, the definition of the vector product must be handled with algebraic rigor to ensure the resulting operator is Hermitian (measurable) and unambiguous.

3.1 Standard Vector Definition

In 3D Euclidean space, we define the operator $\hat{\mathbf{L}}$ as the vector product of the position and momentum operators:

$$\hat{\mathbf{L}} \equiv \hat{\mathbf{r}} \times \hat{\mathbf{p}} = -i\hbar(\mathbf{r} \times \nabla). \tag{14}$$

Using the Levi-Civita pseudotensor ϵ_{ijk}, the components are defined as:

$$\hat{L}_i = \epsilon_{ijk}\hat{x}_j\hat{p}_k, \tag{15}$$

where Einstein summation is implied. This definition ensures that $\hat{\mathbf{L}}$ acts as a vector operator under spatial rotations.

3.2 Geometric Algebra Perspective (Clifford Algebra)

To provide a coordinate-independent definition valid in any dimension, we employ the Geometric Product within the algebra $\mathcal{C}\ell_{3,0}(\mathbb{R})$. The product of the vector operators $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ decomposes into a scalar (symmetric) and bivector (antisymmetric) part:

$$\hat{\mathbf{r}}\hat{\mathbf{p}} = \hat{\mathbf{r}} \cdot \hat{\mathbf{p}} + \hat{\mathbf{r}} \wedge \hat{\mathbf{p}}. \tag{16}$$

The physically significant quantity is the bivector $\hat{\mathcal{L}}$, representing the oriented plane of rotation:

$$\hat{\mathcal{L}} \equiv \hat{\mathbf{r}} \wedge \hat{\mathbf{p}} = \frac{1}{2}(\hat{\mathbf{r}}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{\mathbf{r}}). \tag{17}$$

The standard vector operator $\hat{\mathbf{L}}$ is recovered as the Hodge dual of this bivector. Using the unit pseudoscalar $I = \hat{e}_1\hat{e}_2\hat{e}_3$:

$$\hat{\mathbf{L}} = -iI\hat{\mathcal{L}}. \tag{18}$$

This formulation reveals that spin and orbital angular momentum share the same bivector structure, differing only in the vector space they act upon.

3.3 Hermiticity and Observability

For $\hat{\mathbf{L}}$ to represent a physical observable, it must be Hermitian. We examine the adjoint of the component definition (15):

$$\hat{L}_i^\dagger = (\epsilon_{ijk}\hat{x}_j\hat{p}_k)^\dagger. \tag{19}$$

Using the property $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ and the fact that $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}$ are Hermitian ($\hat{x}_j^\dagger = \hat{x}_j$, $\hat{p}_k^\dagger = \hat{p}_k$) and ϵ_{ijk} is real:

$$\hat{L}_i^\dagger = \epsilon_{ijk}\hat{p}_k^\dagger\hat{x}_j^\dagger = \epsilon_{ijk}\hat{p}_k\hat{x}_j. \tag{20}$$

The non-commutativity of $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}$ is resolved by the canonical commutation relation $[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}$:

$$[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk} \implies \hat{p}_k\hat{x}_j = \hat{x}_j\hat{p}_k - i\hbar\delta_{jk}. \tag{21}$$

Substituting this back into the expression for $\hat{L}_i^\dagger$:

$$\begin{aligned}\hat{L}_i^\dagger &= \epsilon_{ijk}(\hat{x}_j\hat{p}_k - i\hbar\delta_{jk}) \\ &= \epsilon_{ijk}\hat{x}_j\hat{p}_k - i\hbar\epsilon_{ijk}\delta_{jk}.\end{aligned} \tag{22}$$

The second term contains the contraction of the antisymmetric tensor ϵ_{ijk} with the symmetric Kronecker delta δ_{jk}. Since ϵ_{ijk} vanishes if any two indices are equal (which δ_{jk} forces), we have:

$$\epsilon_{ijk}\delta_{jk} = \epsilon_{ijj} = 0. \tag{23}$$

Therefore:

$$\hat{L}_i^\dagger = \epsilon_{ijk}\hat{x}_j\hat{p}_k = \hat{L}_i. \tag{24}$$

The operator is inherently Hermitian, and no symmetrization (e.g., $\frac{1}{2}(\mathbf{r}\times\mathbf{p} - \mathbf{p}\times\mathbf{r})$) is required because the cross product couples only commuting orthogonal components (e.g., x and p_y).

3.4 Explicit Component Form

Expanding the tensor notation, we obtain the explicit forms used in standard calculations:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \tag{25}$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \tag{26}$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x. \tag{27}$$

These expressions confirm that the quantum angular momentum operator preserves the functional form of its classical counterpart, provided the order of operators is maintained as $\mathbf{r}\times\mathbf{p}$.

3.5 The Lie Algebra $\mathfrak{so}(3)$

The fundamental commutation relations for the components of $\hat{\mathbf{L}}$ define the structure of the rotation group. Computing $[\hat{L}_x, \hat{L}_y]$:

$$\begin{aligned}[\hat{L}_x, \hat{L}_y] &= [yp_z - zp_y, zp_x - xp_z] \\ &= [yp_z, zp_x] + [-zp_y, -xp_z] \quad \text{(other terms commute)} \\ &= yp_x[p_z, z] + xp_y[z, p_z] \\ &= yp_x(-i\hbar) + xp_y(i\hbar) \\ &= i\hbar(xp_y - yp_x) = i\hbar\hat{L}_z.\end{aligned} \tag{28}$$

Generalizing this result yields the Lie algebra of $\mathfrak{so}(3)$:

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k. \tag{29}$$

This closure relation confirms that $\hat{\mathbf{L}}$ generates the group of spatial rotations SO(3).

4 Coordinate Representation of Angular Momentum

We establish the representation of the abstract operator $\hat{\mathbf{L}}$ acting on the Hilbert space of square-integrable functions $L^2(\mathbb{R}^3)$.

4.1 Cartesian Representation

In the position basis $\{|\mathbf{r}\rangle\}$, the momentum operator acts as the gradient generator $\hat{\mathbf{p}} \to -i\hbar\nabla$. Consequently, the angular momentum operator $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ becomes a differential operator:

$$\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla). \tag{30}$$

Expanding the cross product yields the explicit differential forms for the generators of rotations about the Cartesian axes:

$$\begin{aligned}
\hat{L}_x &= -i\hbar \left(y\partial_z - z\partial_y \right), \\
\hat{L}_y &= -i\hbar \left(z\partial_x - x\partial_z \right), \\
\hat{L}_z &= -i\hbar \left(x\partial_y - y\partial_x \right).
\end{aligned} \tag{31}$$

While valid, these coupled expressions obscure the separation of radial and angular dynamics, necessitating a transition to spherical coordinates.

4.2 Spherical Coordinate Transformation

We employ the spherical coordinates (r, θ, ϕ) where $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, and $z = r\cos\theta$.

4.2.1 The Generator of Azimuthal Rotations ($\hat{L}_z$)

The operator $\hat{L}_z$ generates rotations about the z-axis (azimuthal symmetry). Using the chain rule for the partial derivative ∂_ϕ:

$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\partial_x + \frac{\partial y}{\partial\phi}\partial_y + \frac{\partial z}{\partial\phi}\partial_z = -y\partial_x + x\partial_y. \tag{32}$$

Comparing this to the Cartesian form of $\hat{L}_z$, we identify the identity:

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}. \tag{33}$$

This simple form confirms that $\hat{L}_z$ is the canonical conjugate momentum to the azimuthal angle ϕ.

To derive the representation of the total angular momentum squared, we utilize the geometric decomposition of the gradient operator. In spherical coordinates, ∇ separates into radial and tangential components:

$$\nabla = \hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \frac{1}{r}\nabla_\Omega, \tag{34}$$

where ∇_Ω denotes the angular gradient on the unit sphere S^2. Substituting this into the definition $\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla)$ and noting that $\mathbf{r} = r\hat{\mathbf{e}}_r$:

$$\hat{\mathbf{L}} = -i\hbar\, r\hat{\mathbf{e}}_r \times \left(\hat{\mathbf{e}}_r \partial_r + \frac{1}{r}\nabla_\Omega\right). \tag{35}$$

Since $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_r = 0$, the radial derivative vanishes, proving that angular momentum is strictly a transverse operator (independent of radial motion):

$$\hat{\mathbf{L}} = -i\hbar(\hat{\mathbf{e}}_r \times \nabla_\Omega). \tag{36}$$

4.2.2 The Casimir Invariant $\hat{L}^2$

To derive the squared total angular momentum $\hat{L}^2 = \hat{\mathbf{L}} \cdot \hat{\mathbf{L}}$, we avoid the tedious squaring of Cartesian components and instead use the vector properties of the Gradient in spherical coordinates.

The gradient operator ∇ is expressed in the spherical orthonormal basis $(\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi)$ as:

$$\nabla = \hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}. \tag{37}$$

Using $\mathbf{r} = r\hat{\mathbf{e}}_r$ and calculating the cross product $\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla)$:

$$\hat{\mathbf{L}} = -i\hbar \left[r\hat{\mathbf{e}}_r \times \left(\hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi} \right) \right]. \tag{38}$$

Using the orthogonality relations $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_r = 0$, $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\phi$, and $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi = -\hat{\mathbf{e}}_\theta$:

$$\begin{aligned}\hat{\mathbf{L}} &= -i\hbar \left[\left(r\frac{1}{r}\right)(\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta)\frac{\partial}{\partial \theta} + \left(r\frac{1}{r\sin\theta}\right)(\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi)\frac{\partial}{\partial \phi} \right] \\ &= -i\hbar \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \hat{\mathbf{e}}_\theta \frac{1}{\sin\theta}\frac{\partial}{\partial \phi} \right). \end{aligned} \tag{39}$$

We apply the vector operator $\hat{\mathbf{L}}$ from Eq. (39) to itself. Extreme care must be taken because the unit vectors $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\phi$ depend on the coordinates, so derivatives acting on them are non-zero.

$$\hat{L}^2 = (-i\hbar)^2 \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin\theta}\frac{\partial}{\partial \phi} \right) \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin\theta}\frac{\partial}{\partial \phi} \right). \tag{40}$$

Expanding the dot product:

$$\frac{\hat{L}^2}{-\hbar^2} = \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta}\left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta}\right)}_{\text{Term 1}} - \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta}\left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta}\frac{\partial}{\partial \phi}\right)}_{\text{Term 2}} - \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi}\left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta}\right)}_{\text{Term 3}} + \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi}\left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta}\frac{\partial}{\partial \phi}\right)}_{\text{Term 4}}. \tag{41}$$

Required Derivatives of Unit Vectors:

- $\partial_\theta \hat{\mathbf{e}}_\phi = 0$.
- $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$ (Orthogonal to $\hat{\mathbf{e}}_\phi$, so dot product vanishes).
- $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin\theta \hat{\mathbf{e}}_r - \cos\theta \hat{\mathbf{e}}_\theta$.
- $\partial_\phi \hat{\mathbf{e}}_\theta = \cos\theta \hat{\mathbf{e}}_\phi$.

Evaluating Terms:

Term 1: Since $\partial_\theta \hat{\mathbf{e}}_\phi = 0$:

$$\hat{\mathbf{e}}_\phi \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \theta^2}\right) = \frac{\partial^2}{\partial \theta^2}.$$

Term 2: Contains $\hat{\mathbf{e}}_\phi \cdot \partial_\theta \hat{\mathbf{e}}_\theta$. Since $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$, the dot product with $\hat{\mathbf{e}}_\phi$ is 0. Cross terms vanish here.

Term 3:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \left((\partial_\phi \hat{\mathbf{e}}_\phi)\frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \phi \partial \theta}\right).$$

Using $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin\theta \hat{\mathbf{e}}_r - \cos\theta \hat{\mathbf{e}}_\theta$:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \left(-\cos\theta \hat{\mathbf{e}}_\theta \frac{\partial}{\partial \theta}\right) = -\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial \theta} = -\cot\theta \frac{\partial}{\partial \theta}.$$

Term 4:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi}\left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta}\frac{\partial}{\partial \phi}\right).$$

The derivative acts on $\hat{\mathbf{e}}_\theta$ and the function. However, $\partial_\phi \hat{\mathbf{e}}_\theta = \cos\theta \hat{\mathbf{e}}_\phi$, which is orthogonal to the pre-factor $\hat{\mathbf{e}}_\theta$. Thus, only the derivative on the scalar function survives:

$$\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial \phi^2}.$$

Summing the non-vanishing contributions:

$$\frac{\hat{L}^2}{-\hbar^2} = \frac{\partial^2}{\partial \theta^2} + \cot\theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial \phi^2}. \tag{42}$$

Recognizing that $\frac{\partial^2}{\partial \theta^2} + \cot\theta \frac{\partial}{\partial \theta} = \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial}{\partial \theta}\right)$, we arrive at the final expression:

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right]. \tag{43}$$

5 Rotational Symmetry and Conservation Laws

We establish the connection between the algebraic generators derived in the previous section and the geometric symmetries of the physical system. This quantizes the classical Noether's theorem: the invariance of the Hamiltonian under spatial rotations implies the conservation of angular momentum.

5.1 The Rotation Operator

5.1.1 Exponential Map

Rotations in the Hilbert space are implemented by the unitary representation of the group SO(3). For a rotation characterized by the vector $\boldsymbol{\theta} = \theta \mathbf{n}$ (rotation by angle θ about axis $\mathbf{n}$), the unitary operator is generated by $\hat{\mathbf{L}}$:

$$\hat{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{\hbar} \boldsymbol{\theta} \cdot \hat{\mathbf{L}} \right). \tag{44}$$

Since $\hat{\mathbf{L}}$ is Hermitian, $\hat{R}$ is Unitary ($\hat{R}^\dagger \hat{R} = \mathbb{I}$), preserving the normalization and probability currents of the state.

5.1.2 Transformation of Vector Operators

A vector operator $\hat{\mathbf{V}}$ (such as position $\hat{\mathbf{r}}$ or momentum $\hat{\mathbf{p}}$) is defined by its transformation properties under the adjoint action of the rotation group. Using the Baker-Campbell-Hausdorff lemma, the components transform as:

$$\hat{R}^\dagger(\boldsymbol{\theta}) \hat{V}_i \hat{R}(\boldsymbol{\theta}) = \sum_j R_{ij}(\boldsymbol{\theta}) \hat{V}_j, \tag{45}$$

where R_{ij} are the components of the classical 3×3 rotation matrix. Specifically for position eigenkets, this implies:

$$\hat{R}(\boldsymbol{\theta}) \left| \mathbf{r} \right\rangle = \left| R\mathbf{r} \right\rangle. \tag{46}$$

5.1.3 Transformation of Scalar Fields

The action of the rotation operator on a scalar wavefunction $\psi(\mathbf{r})$ is derived from the requirement that the physical state remains invariant under a change of frame.

$$\psi'(\mathbf{r}) = \left\langle \mathbf{r} \right\rangle \psi' = \left\langle \mathbf{r} \right| \hat{R}(\boldsymbol{\theta}) \left| \psi \right\rangle. \tag{47}$$

Inserting the identity $\int d^3r' \left|\mathbf{r}'\right\rangle \left\langle\mathbf{r}'\right| = \mathbb{I}$ and using $\langle\mathbf{r}\rangle\,\mathbf{r}' = \delta(\mathbf{r}-\mathbf{r}')$:

$$\psi'(\mathbf{r}) = \psi(R^{-1}\mathbf{r}). \tag{48}$$

For an infinitesimal rotation $d\boldsymbol{\theta}$, the inverse rotation is $\mathbf{r} - d\boldsymbol{\theta}\times\mathbf{r}$. Expanding to first order:

$$\begin{aligned}\psi(\mathbf{r} - d\boldsymbol{\theta}\times\mathbf{r}) &\approx \psi(\mathbf{r}) - (d\boldsymbol{\theta}\times\mathbf{r})\cdot\nabla\psi(\mathbf{r}) \\ &= \psi(\mathbf{r}) - \frac{i}{\hbar}d\boldsymbol{\theta}\cdot(\mathbf{r}\times -i\hbar\nabla)\psi(\mathbf{r}) \\ &= \left(\mathbb{I} - \frac{i}{\hbar}d\boldsymbol{\theta}\cdot\hat{\mathbf{L}}\right)\psi(\mathbf{r}). \end{aligned} \tag{49}$$

This recovers the generator definition $\hat{\mathbf{L}} = \hat{\mathbf{r}}\times\hat{\mathbf{p}}$ consistently from the field transformation perspective.

5.2 Conservation Laws

5.2.1 Scalar Operators

An operator $\hat{S}$ is termed a *scalar operator* if it is invariant under rotations, which implies it commutes with the generators:

$$[\hat{L}_i, \hat{S}] = 0. \tag{50}$$

The dot product of any two vector operators $\hat{\mathbf{A}}\cdot\hat{\mathbf{B}}$ forms a scalar operator. We verify this generally using the tensor commutator $[\hat{L}_i, \hat{A}_j] = i\hbar\epsilon_{ijk}\hat{A}_k$:

$$\begin{aligned}[\hat{L}_i, \hat{\mathbf{A}}\cdot\hat{\mathbf{B}}] &= [\hat{L}_i, \hat{A}_j\hat{B}_j] \\ &= [\hat{L}_i, \hat{A}_j]\hat{B}_j + \hat{A}_j[\hat{L}_i, \hat{B}_j] \\ &= i\hbar\epsilon_{ijk}\hat{A}_k\hat{B}_j + i\hbar\epsilon_{ijk}\hat{A}_j\hat{B}_k. \end{aligned} \tag{51}$$

Summing over j, k, the term $\hat{A}_k\hat{B}_j$ is symmetric (assuming $[\hat{A}, \hat{B}] = 0$ or similar vector types) while ϵ_{ijk} is antisymmetric. Thus, the sum vanishes.

5.2.2 Invariance of the Hamiltonian

The free particle Hamiltonian $\hat{H} = \frac{\hat{\mathbf{p}}\cdot\hat{\mathbf{p}}}{2m}$ is a specific instance of a scalar operator (contraction of vector $\hat{\mathbf{p}}$ with itself). Therefore:

$$[\hat{L}_i, \hat{H}] = \frac{1}{2m}[\hat{L}_i, \hat{\mathbf{p}}^2] = 0. \tag{52}$$

5.2.3 Time Independence

Applying the Heisenberg equation of motion (or Ehrenfest's theorem):

$$\frac{d}{dt}\hat{\mathbf{L}}_H(t) = \frac{1}{i\hbar}[\hat{\mathbf{L}}, \hat{H}] + \frac{\partial \hat{\mathbf{L}}}{\partial t} = 0. \tag{53}$$

This confirms that for an isotropic free particle, angular momentum is a strict constant of motion, and the states may be simultaneously labeled by energy and angular momentum quantum numbers.

5.2.4 Classical Correspondence and Isomorphism

The conservation law established above is the quantum manifestation of the classical rotational invariance. Classically, the time evolution of angular momentum is governed by the Poisson bracket with the Hamiltonian:

$$\frac{d\mathbf{L}_{cl}}{dt} = \{\mathbf{L}_{cl}, H_{cl}\}_{PB}. \tag{54}$$

For a free particle, the absence of external torque implies $\{\mathbf{L}_{cl}, H_{cl}\}_{PB} = 0$. The quantum result is structurally identical, following the Dirac quantization correspondence:

$$\lim_{\hbar \to 0} \frac{1}{i\hbar}[\hat{\mathbf{L}}, \hat{H}] \longrightarrow \{\mathbf{L}_{cl}, H_{cl}\}_{PB}. \tag{55}$$

This confirms that the role of $\hat{\mathbf{L}}$ is twofold and consistent across regimes: it is simultaneously the physical observable corresponding to classical angular momentum and the infinitesimal generator of the rotation group SO(3) imposed by the isotropy of the vacuum.

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